Temporal Structure Governs Hierarchical Lattice Responses under External Driving

Abstract

We investigate what an external drive's temporal structure changes inside a kagome XY lattice, beyond the waveform discrimination established previously (Paper III). Nine representative signals — sine, square, white noise, random telegraph, pink noise, brown noise, an Ornstein-Uhlenbeck process, a chaotic logistic map, and a sparse pulse train — were driven into the lattice at matched injected power, and each input's own temporal features (autocorrelation time, spectral slope, permutation entropy, smoothness, sparsity) were quantified alongside the lattice's dynamical, informational, and morphological response.

Principal component analysis and hierarchical clustering of the resulting response profiles show that the nine signals organize spontaneously along an axis matching their input autocorrelation time, without this axis being imposed by the analysis; the same ordering is directly confirmed by a strong correlation between input autocorrelation time and response effect size (r=-0.85). Across a driving-power sweep (n=50 seeds) and an independent spatially-resolved comparison (uniform versus checkerboard driving at fixed power), the lattice's observables consistently separate into two groups. Intensity statistics — absorbed energy, vortex-birth rate, the information field's peak and mean, and how these quantities fluctuate over time — respond readily to both manipulations. Spatial-organization statistics — the field's correlation length, its number of connected components, and its largest-component fraction — remain comparatively robust to both.

An earlier analysis based on fewer seeds supported a three-tier, driving-power-ordered hierarchy; increasing the sample size closed the apparent gap between two of the three tiers entirely and showed that the third (temporal organization) activates according to descriptor type rather than at a single shared power. We report this revision transparently, since it clarifies that the two-tier, descriptor-type split — not the finer power-ordering — is the structure that remained stable under additional sampling and independent spatial-pattern manipulation.

1. Introduction

Paper III of this series established that a driven kagome XY lattice discriminates the temporal organization of an external perturbation, independent of the energy that perturbation injects: waveforms carrying statistically indistinguishable absorbed energy nonetheless produce clearly separated transfer entropy. That result answers whether the lattice distinguishes waveforms. It does not address a related but distinct question: when a waveform's temporal structure does register in the lattice, what, specifically, does it change?

This question can be split further. A driven lattice supports several qualitatively different classes of observable: instantaneous dynamical quantities (absorbed energy, vortex-birth rate), the intensity of the induced information field (its spatial peak, Smax), and the temporal organization of that field (how its fluctuations are structured over time). It is not obvious a priori that these three classes respond to driving in the same way, at the same driving strength, or even in the same order. If they do not, then "the lattice responds to temporal structure" is an incomplete statement — the more precise statement is that specific layers of the lattice's response respond at specific driving strengths.

We address this in two steps. First, we ask whether nine signals spanning a broad range of temporal structure — from memoryless white noise to the extreme long-memory limit of brown noise, covering four decades of autocorrelation time — organize the lattice's response space along recognizable axes, using representative parameter values for each signal family rather than a parameter sweep (Section 2.1). Second, using three of these signals chosen to span the resulting response landscape, we sweep driving power and ask, separately for each observable class, at what power that class first becomes statistically distinguishable from its unperturbed baseline. Section 2 describes the model and methods, including the operational, model-independent definition of this threshold. Section 3 presents the results. Section 4 discusses their interpretation and relation to Papers I–III.

2. Model and Methods

2.1 Nine representative signals

As in Papers I–III, we use the driven kagome XY lattice $H(t) = -K(t)\,\Sigma\langle ij\rangle \cos(\varphi_i - \varphi_j)$ with $K(t) = K_0 + \delta K(t)$, $N=3L^2$ sites, $L=48$, evolved by the Metropolis algorithm at $T=0.9$, $K_0=0.7$ (the BKT-adjacent regime established in Paper I).

Following the principle established in Paper III — that a single question should vary one axis at a time — this study uses one representative parameter value per signal family, rather than sweeping each family's internal parameters. The nine signals and their fixed parameters are: sine (period=64 sweeps), square (period=64), white noise, random telegraph (mean dwell=40 sweeps), pink noise (1/f spectrum), brown noise ($1/f^2$ spectrum), an Ornstein-Uhlenbeck process (τ=40 sweeps), a chaotic logistic map (r=4.0), and a sparse pulse train (duty cycle=0.05). All nine are normalized to identical injected power $P=\langle\delta K^2\rangle=0.01$ for the baseline comparison (Section 3.1–3.2); the power-sweep stage (Section 3.3–3.5) varies P at fixed waveform for three of these signals.

For each signal, we additionally quantify eight input-side features directly from the driving time series, independent of the lattice's response: Shannon entropy, spectral entropy, spectral slope β (PSD~1/f^β), a Hurst exponent (via detrended fluctuation analysis), an integrated autocorrelation time (Sokal estimator), permutation entropy, a smoothness measure (mean normalized $|\Delta x|$), and a sparsity duty cycle. These allow response patterns to be related to quantitative input properties rather than to waveform identity alone.

2.2 Response observables: intensity and spatial organization

At each measurement step we record the same core observables as Papers I–III (absorbed energy, vortex-birth rate, local restoring strength drop, collective information density √(MI), helicity modulus), together with morphological descriptors of the information field I(x,y) computed at every measurement step: its spatial mean, maximum (Smax), standard deviation, skewness, kurtosis, correlation length, number of connected components, and largest-component fraction (Paper II’s H-map construction, R_LOCAL=3).

These observables are grouped into two classes, examined throughout this paper. Intensity statistics comprise the instantaneous, energy-linked quantities (absorbed energy, vortex-birth rate, drop, helicity, summarized by their baseline-vs-driven effect size, Hedges' g), the information field's spatial mean and peak (Smax), and the time-resolved fluctuation of each morphological descriptor within the driven period — its temporal standard deviation, integrated autocorrelation time, and coefficient of variation, when that descriptor is itself an intensity-type quantity (mean, maximum, or standard deviation). Spatial-organization statistics comprise the information field's correlation length, its number of connected components, and its largest-component fraction, together with the time-resolved fluctuation of the latter two connectivity descriptors. This grouping is adopted based on the results of Sections 3.1–3.6, rather than assumed in advance; Section 2 here only defines the observables, deferring their classification to Results.

2.3 Operational threshold: a model-independent hierarchy metric

To compare when each of the three response classes first becomes detectable as driving power increases, we define an operational threshold that makes no assumption about the functional form of the power-response relationship. For a given observable and a given waveform, the lowest power tested serves as baseline; for every higher power tested, the observable's distribution across seeds is compared against the baseline distribution (Mann-Whitney U), and the resulting p-values are Holm-Bonferroni corrected across all comparisons for that observable. The threshold is the lowest power at which the corrected p-value falls below 0.05; if no such power exists within the tested range, the threshold is undefined (not detected).

This definition is deliberately not tied to a specific curve shape. A sigmoid fit (yielding a half-maximum power, P50) was also examined as a secondary, descriptive reference where the response curve visibly saturates (Section 3.3); it is not used as the primary hierarchy metric; because it assumes a functional form that not all three response classes need share, and because a sigmoid fit is unstable when a class (dynamics, in this case) shows no flat baseline within the tested power range.

2.4 Statistical testing

Nine-signal comparisons (Section 3.1–3.2, n=10 seeds per signal) use one-way ANOVA and the non-parametric Kruskal-Wallis test, with Kruskal-Wallis effect size ε^2 reported alongside. The driving-power sweep (Section 3.3) uses white noise at eight power levels (0.005, 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.1), n=50 seeds per power level; the temporal-organization scan (Section 3.4) uses all nine signals at three power levels (0.01, 0.05, 0.1), n=30 seeds per (signal, power) combination. The spatially-resolved comparison (Section 3.6) uses n=10 seeds per condition. Input-output relationships (Section 3.1) are assessed by Spearman correlation. All pairwise comparisons following a significant omnibus test use Holm-Bonferroni correction.

2.5 Spatially-resolved driving

Sections 3.1–3.5 use the spatially uniform driving protocol of Papers II–III, $\delta K(t)$, applied identically to every bond. Section 3.6 additionally uses a spatially-resolved protocol, $\delta K(\text{bond}, t)$, which the simulation architecture now supports as a two-dimensional array (sweep × bond) rather than a scalar time series. The checkerboard pattern assigns each bond to one of two sublattices, determined by mapping each bond's real-space midpoint onto the kagome lattice's unit-cell coordinates and taking the parity of their sum; the two sublattices receive the same sine waveform with opposite sign, so that each individual bond still receives exactly the target injected power, and only the spatial arrangement of that power differs from the uniform case.

3. Results

3.1 The input space organizes the lattice's response along a temporal-structure axis

Figure 1 summarizes the nine-signal response landscape from three independent angles. Principal component analysis of the seventeen driven-response descriptors (energetic deltas and effect sizes, information-field morphology) reduces to two components explaining 73.7% and 10.4% of variance (Figure 1A), separating brown, pink, telegraph, sine, and OU noise — the five longest-autocorrelation signals — from white noise, the logistic map, and the pulse train — the three signals with essentially zero autocorrelation — with square noise near the boundary. Hierarchical clustering of the same descriptor space (Figure 1B) recovers this same partition independently, without reference to any input feature, despite {white, logistic, pulse} being generated by unrelated mechanisms (stochastic, deterministic-chaotic, and sparse-discontinuous, respectively).

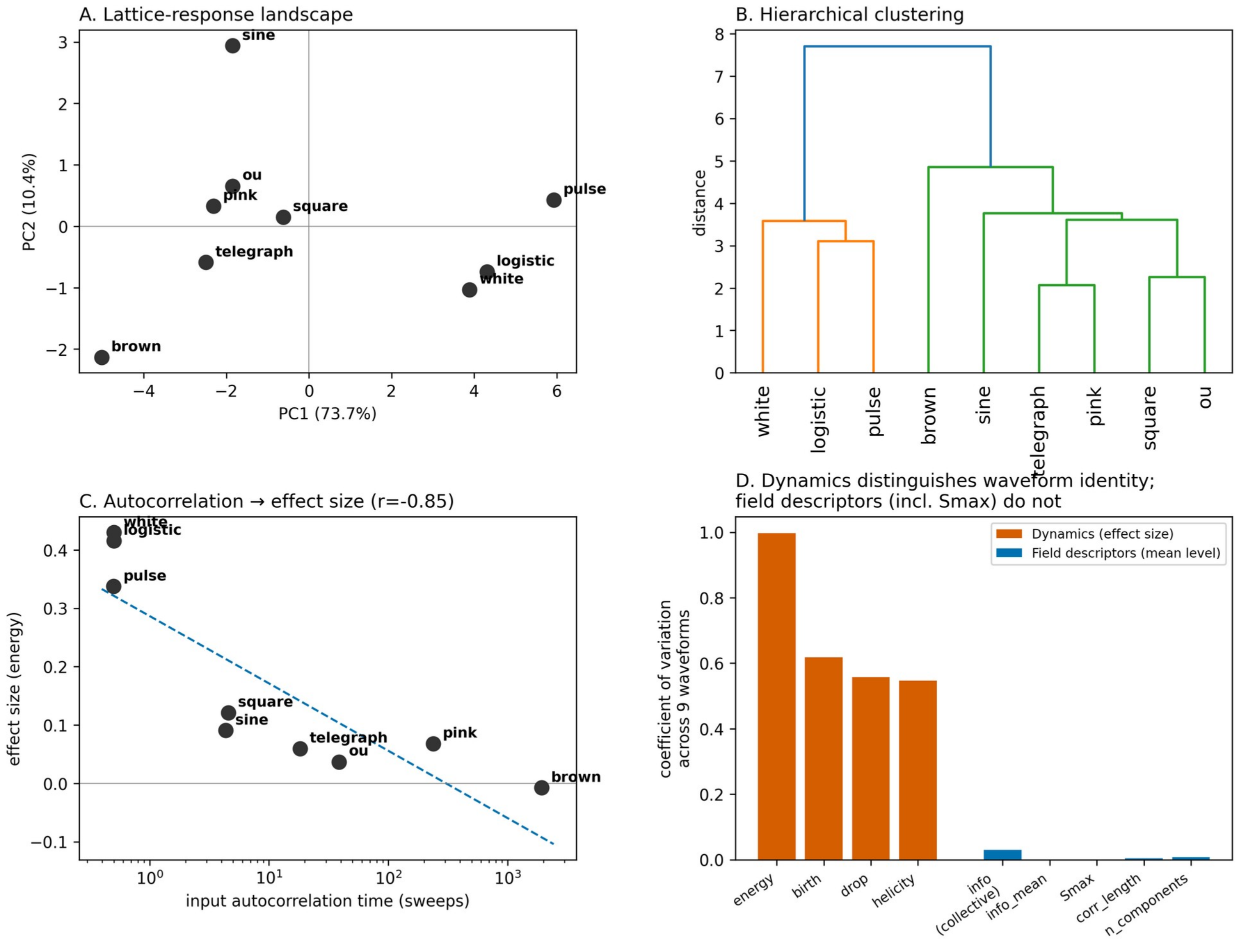

Figure 1. The nine-signal response landscape, viewed four ways. (A) PCA of seventeen response descriptors (energy, birth, drop, helicity deltas and effect sizes; information-field morphology). (B) Hierarchical clustering of the same descriptor space (average linkage, Euclidean distance) separates {white, logistic, pulse} from the remaining six signals. (C) The input's own autocorrelation time (Section 2.1) correlates strongly with response effect size (Spearman r=-0.85), independent of waveform identity or label. (D) Different input waveforms primarily modulate dynamical response metrics, whereas information-field descriptors remain comparatively invariant under fixed driving power.

Figure 1C confirms this pattern directly against a quantitative input feature rather than a categorical waveform label: response effect size correlates with input autocorrelation time across four decades (from ≈0.5 sweeps for white/logistic/pulse to ≈2000 sweeps for brown noise), with Spearman r=-0.85. Brown noise — the extreme long-memory case — lies below a naive log-linear extrapolation of the other eight signals rather than on it, consistent with the relationship saturating near zero rather than continuing linearly; we note this without pursuing the saturating form further, as it is a distinct question from the one addressed here. Figure 1D shows the same fixed-power comparison from a different angle: Dynamics observables (effect sizes for energy, birth, drop, helicity) vary substantially across the nine waveforms (coefficient of variation 0.55–1.0), while field descriptors evaluated at their mean level — including Smax — do not (coefficient of variation below 0.03).

3.2 Representative waveforms span the identified response space

White, telegraph, and sine — the three canonical waveforms used throughout Papers II–III — were selected for the power sweep in part because they already span the structure identified in Section 3.1: white noise anchors the zero-autocorrelation branch, while telegraph and sine occupy separated positions within the long-memory branch (Figure 1A–B), together bracketing a substantial portion of the identified response landscape without requiring a new representative-signal selection.

3.3 Intensity-linked observables converge to a common power threshold

Sweeping driving power at fixed waveform (white; eight power levels, n=50 seeds, Section 2.4) shows that the vortex-birth effect size (Dynamics) and the information-field peak intensity (Smax) become statistically distinguishable from their lowest-power baseline at the same power, 0.01 (operational threshold, Section 2.3). This result revised an earlier analysis based on fewer seeds (n=10), in which Smax appeared to require a higher power (0.02) than Dynamics; that apparent gap closed as the sample size increased, indicating it reflected detection power rather than a genuine separation between the two observables (Section 4.2).

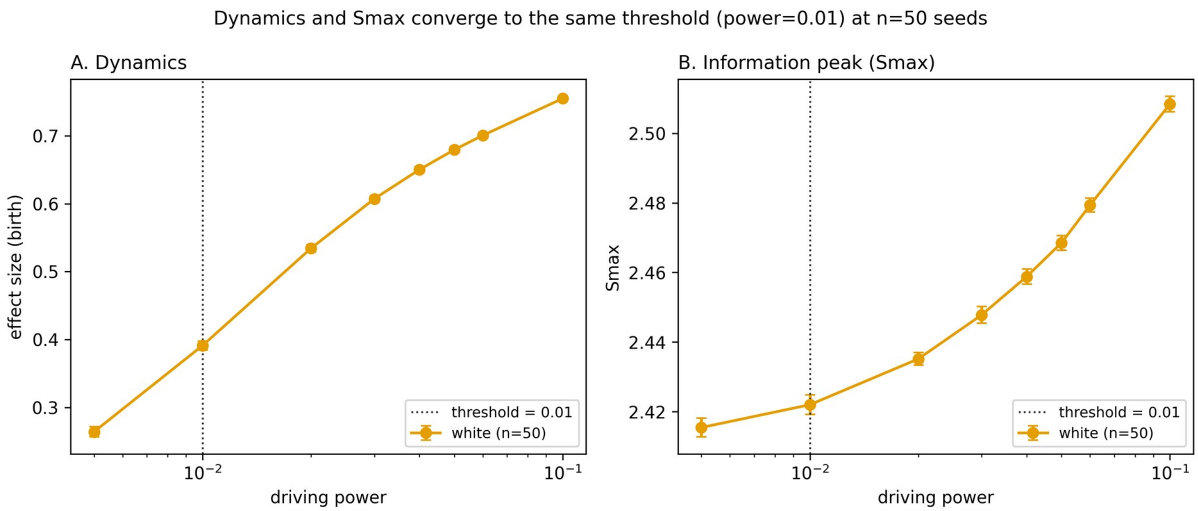

Figure 2. Power-response curves for the vortex-birth effect size (Dynamics, panel A) and the information-field peak intensity (Smax, panel B), for white driving (n=50 seeds). Both reach their operational threshold (Section 2.3) at the same power, 0.01.

3.4 Temporal fluctuation splits by descriptor type, not by driving power

The temporal organization of the information field — how its morphological descriptors fluctuate over time, rather than their mean values — was assessed via eighteen metrics (temporal standard deviation, integrated autocorrelation time, and coefficient of variation, for each of six morphological descriptors; Section 2.2) at three driving powers (0.01, 0.05, 0.1), n=30 seeds, across all nine signals. Of the 162 (signal × metric) combinations tested, 104 (64.2%) reached the operational threshold within the tested range; of those, 76.9% first became significant at power=0.05 and the remaining 23.1% at power=0.1. The remaining 58 combinations (35.8%) showed no statistically detectable change at any power tested.

Critically, whether a given metric activates readily is determined by what it describes, not by which power tier it belongs to. Temporal fluctuation of intensity-linked descriptors — the field's mean, its peak (Smax), and their standard deviation — reached threshold in 8–9 of 9 waveforms, almost always at the lower power tested (0.05). Temporal fluctuation of descriptors of the field's spatial connectivity — the largest-component fraction and the number of connected components — essentially never reached threshold: the largest-component fraction's temporal statistics were undetected in 9 of 9 waveforms at every power tested, and the component-count coefficient of variation in 7 of 9 (Figure 3). Correlation-length statistics fell in between, detected in roughly half of waveforms and split across both powers.

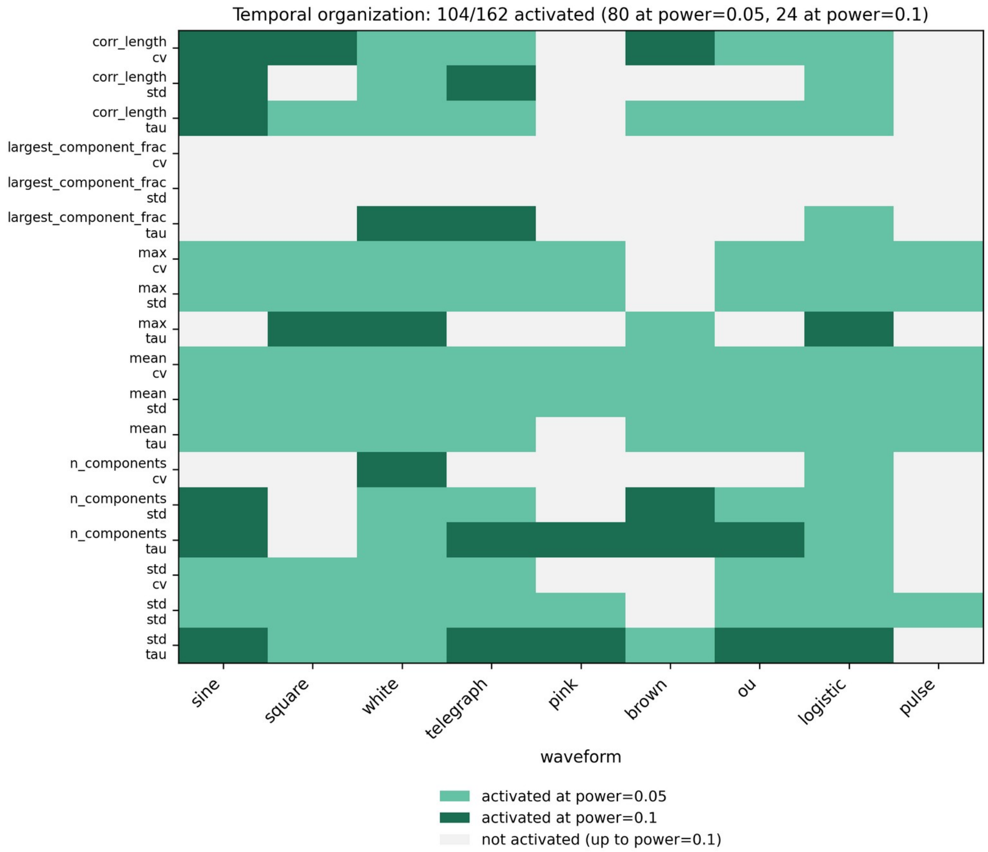

Figure 3. Activation of temporal-organization metrics (18 metrics × 9 signals = 162 combinations, n=30 seeds) across the tested power range. Light green: reached operational threshold at power=0.05. Dark green: reached threshold only at power=0.1. Grey: no detectable change in temporal organization at any tested power. Intensity-linked descriptors (mean, maximum, standard deviation) are concentrated in light green; spatial-connectivity descriptors (largest-component fraction, number of connected components) are concentrated in grey.

3.5 A two-tier structure: intensity versus spatial organization

Sections 3.1–3.4, together with the spatially-resolved comparison in Section 3.6, converge on the same two-tier structure regardless of which axis of driving is varied (power, Section 3.3–3.4; spatial pattern, Section 3.6). Intensity statistics — absorbed energy, vortex-birth rate, the information field's peak and mean, and the temporal fluctuation of these quantities — respond readily: they distinguish waveform identity at fixed power (Section 3.1), activate at the lowest power tested (Section 3.3), and fluctuate detectably over time already at moderate power (Section 3.4). Spatial-organization statistics — the field's correlation length, its number of connected components, and its largest-component fraction, together with the temporal fluctuation of the latter two — are comparatively robust across all of these same manipulations. Figure 4 summarizes this structure.

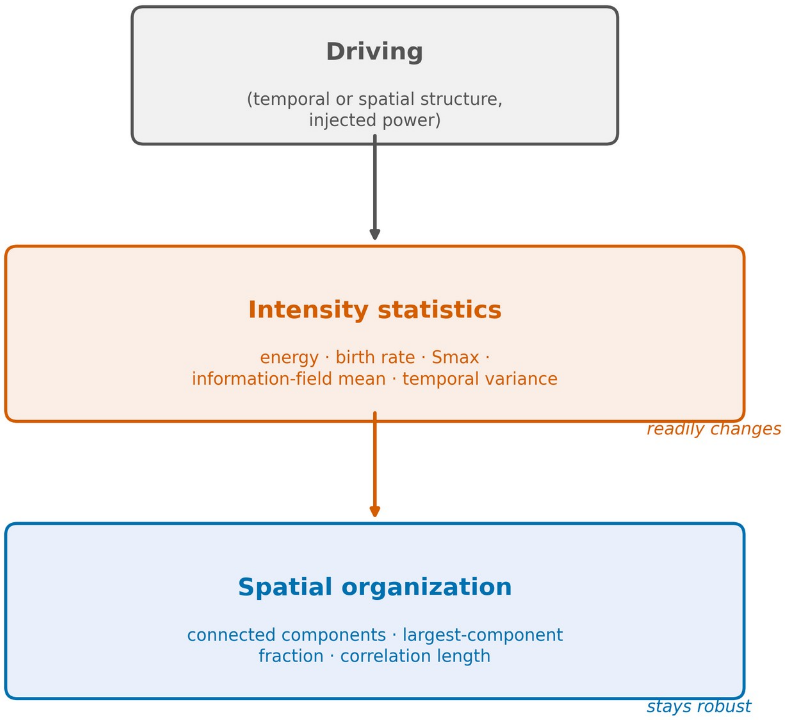

Figure 4. A two-tier structure connects driving to the lattice's response: intensity statistics (energy, birth rate, Smax, information-field mean, and their temporal variance) respond readily to driving, while spatial-organization statistics (connected components, largest-component fraction, correlation length) remain comparatively robust across the power and spatial-pattern manipulations examined in this paper.

3.6 The same split appears along a spatial-pattern axis

As an independent check on this two-tier structure, we compared two spatially-resolved driving protocols at fixed temporal structure and power (sine, period=64, power=0.01, n=10 seeds): a spatially uniform drive, identical to that used throughout Sections 3.1–3.5, and a checkerboard pattern in which the same sine signal is applied with opposite sign to alternating bonds of the kagome unit cell (Section 2.5). Of sixteen observables compared (Dynamics deltas and effect sizes, six morphology means), eight reached significance after Holm-Bonferroni correction — and again, every significant observable was an intensity statistic (absorbed energy, drop, birth rate, and the information field's mean, together with their effect sizes), while every spatial-organization statistic (Smax, correlation length, number of connected components, largest-component fraction) was statistically indistinguishable between uniform and checkerboard driving. Absorbed energy's effect size was markedly larger under checkerboard driving than under uniform driving at the same power (0.57 versus 0.09), and the sign of the drop delta reversed between the two conditions. This independent manipulation — varying spatial pattern rather than power — recovers the same intensity/spatial-organization split identified in Sections 3.3–3.4.

4. Discussion

4.1 Intensity responds; spatial organization resists

The central finding of this paper is that a driven kagome lattice's observables separate into two groups with markedly different sensitivity to driving, and that this separation is robust across two independent ways of varying the drive: increasing its power (Sections 3.3–3.4) and rearranging it spatially at fixed power (Section

3.6). Intensity statistics — absorbed energy, vortex-birth rate, and the information field's peak and mean, together with how these quantities fluctuate over time — respond readily to both manipulations. Spatial-organization statistics — the field's correlation length, its number of connected components, and its largest-component fraction — are comparatively robust to both. We describe this as intensity responding while spatial organization resists, rather than as a strict hierarchy of driving-power thresholds: the earlier working hypothesis, that Dynamics, Smax, and temporal organization form three sequentially-activated tiers ordered by driving power, did not survive increased sampling (Section 4.2) and is superseded by the present two-tier, descriptor-type-based structure.

This picture is consistent with, and extends, Paper III's finding that the lattice discriminates temporal organization independent of energy: the present results indicate that this discrimination is carried predominantly by intensity statistics, while the field's spatial connectivity remains comparatively insensitive to how the drive is organized in time or space.

4.2 A revision made necessary by re-sampling, reported for transparency

An earlier analysis of this dataset, based on n=10 seeds for the power sweep and n=10 seeds for the temporal-organization scan, supported a three-tier hierarchy in which Dynamics (power≈0.01), Smax (power≈0.02), and temporal organization (power≥0.1, uniformly) activated at successively higher, cleanly separated driving powers. Increasing the sample size (to n=50 for the power sweep and n=30 for the temporal-organization scan, Section 2.4) closed the gap between Dynamics and Smax entirely (both converge to power=0.01) and revealed that the temporal-organization metrics do not activate uniformly at the highest power tested: most (76.9% of those that activate at all) already do so at an intermediate power, and this activation depends on which descriptor is examined — intensity-linked descriptors activate readily, spatial-connectivity descriptors largely do not — rather than on a single shared threshold. We report both the original and revised analyses in this section, rather than silently updating the result, because the direction of the revision (an apparent three-tier ordering collapsing into a two-tier, descriptor-type split under additional sampling) is itself informative: it indicates that the operational threshold (Section 2.3), while model-independent, is still a statement about detectability at a given sample size, and that conclusions about the relative ordering of nearby thresholds should be treated cautiously until confirmed at higher sample sizes, as done here.

The operational threshold retains its value as a model-independent detection criterion (Section 2.3): it does not assume a functional form for the power-response curve, and it is defined identically across effect sizes, raw means, and morphology-derived statistics, which is what allows intensity and spatial-organization statistics to be compared on a common footing despite being measured on entirely different scales. What the present revision changes is not the method but the granularity of claim it supports: individual threshold values close to one another (e.g., 0.01 versus 0.02) should not be treated as evidence of ordering without confirming that the gap is stable under increased sampling, whereas the much larger and qualitatively different gap between intensity statistics (threshold ≤0.01–0.05) and spatial-organization statistics (mostly undetected up to 0.1) proved stable under the same re-sampling and is the structure we report as the paper's central result.

4.3 Relation to Papers I–III

Paper I characterized the lattice's equilibrium information structure; Paper II identified a local energetic mechanism (drop) linking energy to vortex formation while showing that collective information remains an emergent, non-reducible property; Paper III showed that the lattice's nonequilibrium response discriminates a drive's temporal organization independent of its energy. The present paper (Paper IV-A) extends this sequence from equilibrium structure (Paper I), through a local mechanism (Paper II) and waveform discrimination (Paper III), to a structural split within the driven response itself: intensity statistics, which include the energetic and vortex-related quantities studied in Papers II–III, are readily perturbed by the drive, while the field's spatial organization — a property not separately examined in Papers I–III — is comparatively robust to it.

4.4 Scope and limitations

Two observations encountered during this study are noted but deliberately excluded from the results above, as they raise distinct questions from the one addressed here. First, a fine-grained sine-period scan (Supplementary analysis) identified a statistically robust, reproducible dip in transfer entropy centered near period≈50 sweeps (width ≈12 sweeps), present for sine but not for square-wave driving at the same periods; whether this reflects a genuine lattice-intrinsic timescale is left for dedicated follow-up work. Second, brown noise — the longest-autocorrelation signal examined — uniquely produced a small but statistically significant negative Δenergy; accompanying vortex-birth and helicity-modulus deltas remained perturbed in the same direction as the other eight signals (just to a lesser degree), providing no independent evidence of active lattice ordering, so we do not interpret this as a distinct cooling mechanism.

More broadly, the spatially-resolved comparison in Section 3.6 used a single spatial pattern (checkerboard) and a single waveform/power combination; whether the intensity/spatial-organization split holds for other spatial patterns (chiral, spiral, or vortex-like driving) is a natural next question, left for a dedicated follow-up study now that the underlying simulation architecture supports spatially-resolved driving (Section 2.5).

5. Conclusions

1. Across both driving-power and spatial-pattern manipulations, the lattice's observables separate into two groups with markedly different sensitivity: intensity statistics (absorbed energy, vortex-birth rate, the information field's peak and mean, and their temporal fluctuation) respond readily, while spatial-organization statistics (correlation length, number of connected components, largest-component fraction) are comparatively robust.
2. This two-tier structure was confirmed independently along two separate axes of variation — increasing driving power at fixed spatial pattern (Sections 3.3–3.4), and rearranging the drive spatially at fixed power (Section 3.6) — with the same descriptor-type split recovered in both cases.
3. An earlier three-tier account of this structure, based on a smaller sample, did not survive re-sampling: the apparent gap between Dynamics and Smax closed entirely at higher sample size, and temporal-organization metrics were found to activate according to descriptor type rather than uniformly at a single driving power. This revision is reported transparently (Section 4.2) because it clarifies which claims are robust (the intensity/spatial-organization split) and which were sampling-dependent (the fine-grained ordering of individual thresholds).

Collectively, these results indicate that a driven lattice's response to temporal and spatial structure is organized not as a sequence of driving-power thresholds, but as a stable division between quantities that track how strongly the lattice is being driven (intensity) and quantities that track how the resulting information is spatially organized (connectivity) — with the latter substantially more robust to variation in the drive than the former, across every manipulation examined in this paper.

Acknowledgments

This work was carried out independently, without institutional affiliation. Large language models (Claude, ChatGPT) were used as auxiliary tools. All simulation design, data interpretation, and conclusions reflect the author's own judgment.

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